

Math728J – MPI Homework 2

1. Write a MPI program to evaluate the definite integral

$$\int_a^b 4x^5 + x^3 + 5x \, dx$$

using trapezoid rule with $[a, b]$ divided into N subregions. Read a, b and N from process 0 and the final result is also stored on process 0. Your program should be general for P processes. Let $a = 0.0$ and $b = 10.0$, print out your results with $N = 10^2, 10^3, 10^4, 10^5$ with $P = 4$. Also print out the timing results and plot the speedup curves for $N = 10^5$ with $P = 1, 2, 4, 8$.

2. Create a $N \times N$ square matrix $\mathbf{A}$ and a vector $\mathbf{x} \in \mathbb{R}^N$ with random elements whose values are within $[0, 1]$. Write a MPI program to evaluate $\mathbf{y} = \mathbf{A}\mathbf{x}$. Assume that initially $\mathbf{A}$ and $\mathbf{x}$ are stored on process 0. Partition the $\mathbf{A}$ into P block row-wise and scatter them across all processes. Store the final result $\mathbf{y}$ on process 0. Verify your parallel result $\mathbf{y}$ by comparing them with the sequential result $\mathbf{y}^*$ obtained by executing $\mathbf{y}^* = \mathbf{A}\mathbf{x}$ only on process 0. Describe your algorithm. Set $N = 1024$. Print out the $\|\mathbf{y} - \mathbf{y}^*\|_2$ from process 0. Print out the times and plot the speedup curves for $P = 1, 2, 4, 8$.
2. Create two $N \times N$ square matrices $\mathbf{A}$ and $\mathbf{B}$ with random elements whose values are within $[0, 1]$. Write a MPI program to evaluate $\mathbf{C} = \mathbf{A}\mathbf{B}$. Assume that initially $\mathbf{A}$ and $\mathbf{B}$ are partitioned into P blocks row-wise and stored independently across all processes. Store the final result $\mathbf{C}$ in blocks row-wise across all processes. Describe your algorithm. Print out the times and plot the speedup curves for $N = 512$ with $P = 1, 2, 4, 8$.